YASH KUMAR

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SUMMARY

B.Tech AI/ML student (3rd year, CGPA 8.5/10) with hands-on experience building and deploying production-grade ML and LLM systems: YOLOv8 computer vision pipelines (94.6% mAP), agentic RAG with hybrid retrieval and LangGraph routing, and LLM agents on GCP using LangChain and Google ADK. Interned in applied search ML (FlyRank) and ad-tech (Cloven Media). Delivered 3 end-to-end projects spanning model training, FastAPI backends, and cloud deployment.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning **2024 – 2028 (Expected)**
University School of Automation & Robotics, GGSIPU EDC | CGPA: 8.5 / 10 (4 semesters) Delhi, India
Relevant Coursework: Machine Learning · Deep Learning · Computer Vision · Data Science · DSA · DBMS · OS · OOPs

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL
ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, YOLOv8, OpenCV
AI & LLMs: LangChain, LangGraph, RAG (hybrid vector + BM25), ChromaDB, Google ADK, Gemini, LLaMA-3, Groq
Backend & DB: FastAPI, Flask, Node.js, Express.js, PostgreSQL, MongoDB, Firebase, DuckDB
Frontend: React.js, Next.js, Tailwind CSS, Streamlit
Cloud & DevOps: Google Cloud Platform (GCP), Cloud Run, Docker, Git, Vercel, Render, Linux

EXPERIENCE

ML Intern — FlyRank **Jul 2026 – Aug 2026**
Applied Search Intelligence — Google Search Ranking & Discoverability *Remote*

- Owned the Refresh / Content Opportunity Scoring lane: built an end-to-end pipeline on FlyRank's production search-ranking warehouse (DuckDB for feature extraction) with a time-aware train/test split to prevent leakage.
- Trained a Random Forest classifier predicting content growth / decline / review status, lifting F1-macro from a 0.208 baseline to **0.380 (~83% relative gain)**.

Artificial Intelligence Intern — Cloven Media **Jun 2026 – Aug 2026**
Summer Internship Program *Remote*

- Built ML models and data pipelines for ad-exchange / programmatic advertising systems, supporting real-time ad-tech decisioning workloads.

KEY PROJECTS

QueDB — AI-Powered NL-to-SQL Engine | quedb.vercel.app **2025**
FastAPI · React · LangChain · LangGraph · LLaMA-3 70B (Groq) · ChromaDB · PostgreSQL · JWT Auth

- Engineered a full-stack AI query engine converting natural language to executable SQL across PostgreSQL, MySQL, and SQLite using LLaMA-3 70B, enabling non-technical users to query databases without writing SQL.
- Built a schema-aware RAG pipeline (ChromaDB) with a self-correcting retry loop for failed queries; added a live Schema Explorer with Monaco editor. Deployed on Railway + Vercel.

RepoSage — Production Codebase RAG Agent | reposage-lake.vercel.app **2026**
FastAPI · React · TypeScript · LangGraph · ChromaDB · BM25 · tree-sitter · Groq (LLaMA-3 / GPT-OSS-120B)

- Built a production-deployed RAG agent that answers natural-language questions about any indexed GitHub repository, with every claim cited back to an exact file and line range.
- Indexed repos via tree-sitter AST-aware chunking with hybrid ChromaDB + BM25 retrieval (Reciprocal Rank Fusion); evaluated cross-encoder reranking and deliberately disabled it after finding a domain mismatch.
- Built a LangGraph agentic router, live indexing API, and a React UI with citation chips and pipeline-trace visualization; deployed on Render + Vercel.

Helmet Detection & Traffic Monitoring System | helmet-detection-project.vercel.app **2025**
Python · YOLOv8 · OpenCV · FastAPI · React · TypeScript · Render · Vercel

- Built a real-time helmet compliance detection system using YOLOv8 (**94.6% mAP@0.5**, 73.9% mAP@0.5:0.95, ~5ms inference) with an overlap-based person-motorcycle matching pipeline and multi-stage detection via a FastAPI REST backend.
- Deployed end-to-end: training, FastAPI (Render), React/TypeScript (Vercel), with Swagger docs.

CERTIFICATIONS

Google Cloud Gen AI Academy APAC 2026 — Hack2skill (Cohorts 1 & 2); AI agents on Cloud Run **Apr – Jul 2026**
IBM AI Fundamentals — IBM SkillsBuild **May 2026**
Claude Code 101 — Anthropic

ACHIEVEMENTS & LEADERSHIP

- **Organizing Team Member, FOSS Club, USAR (GGSIPU EDC):** Organized 3+ hackathons, each with 500+ registrations.
- **Smart India Hackathon 2025:** Certificate of Appreciation; cleared University-Level 2 rounds of India's largest national hackathon.
- **CodeX Hackathon:** Built and presented an AI project under competitive hackathon conditions.